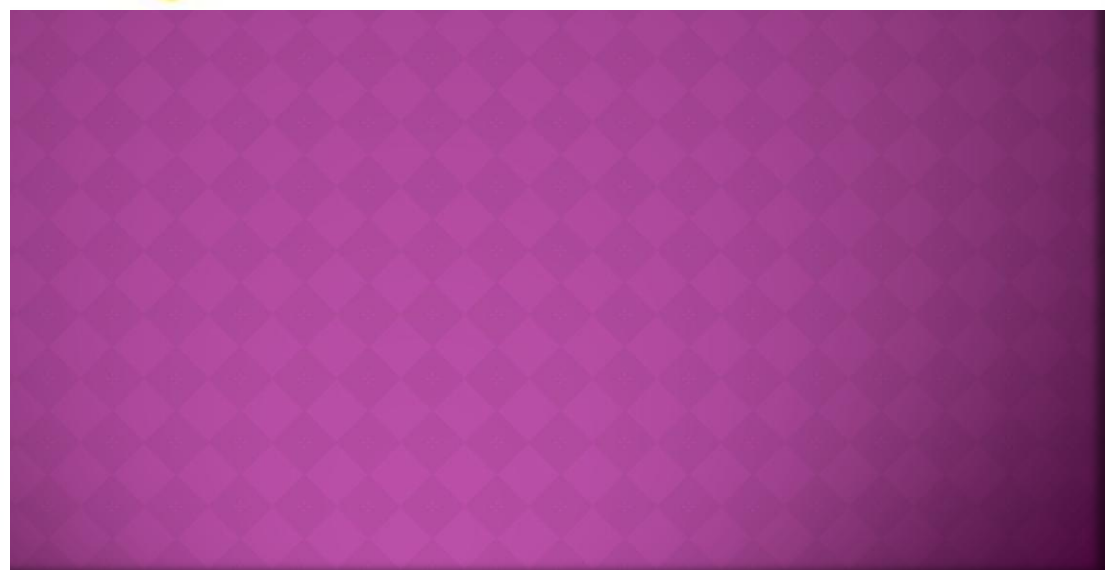




CONTENTS

- ◉ Introduction to waste and types
- ◉ Solid waste
- ◉ Types of solid waste
- ◉ Effects of solid waste
- ◉ Waste management concept
- ◉ Concept of 3R
- ◉ solid waste management
 - storage
 - collection
 - waste handling and transport
 - method of disposal
- ◉ Technology
- ◉ Zero waste system
- ◉ Recommendation

WASTE

- ◉ It is defined as

Waste (also known as garbage, trash, refuse, garbage, junk) is any unwanted or useless materials.

OR

Any materials unused and rejected as worthless or unwanted and “A useless or profile less activity using or expanding or consuming thoughtlessly or carefully.”

TYPES OF WASTE

- ◉ Solid waste
- ◉ Liquid waste
- ◉ Gaseous waste
- ◉ Animal by product(ABPs)
- ◉ Biodegradable waste
- ◉ Chemical waste
- ◉ Commercial waste/
Business waste
- ◉ Biomedical waste
- ◉ Bulky waste

SOLID WASTE

- It is defined as
 - “ non liquid, non-soluble materials ranging from municipal garbage to industrial wastes that contain complex & sometimes hazardous substances”
 - Solid waste also include
 - Garbage
 - Trash
 - Demolition products
 - Sewage treatment residue
 - Dead animals
 - Manure and other discarded material.
- Per capita solid waste out put 0.25-2.5 Kg/day

A photograph showing two people wading in a river, surrounded by many salmon. One person is in the foreground, wearing a blue shirt and a cap, holding a large salmon. Another person is further back, also holding a salmon. The water is filled with many salmon, some of which are being held by the people. The background shows a lush green forest.

-
- A photograph showing two people wading in a river, surrounded by many salmon. One person is in the foreground, wearing a blue shirt and a cap, holding a large salmon. Another person is further back, also holding a salmon. The water is filled with many salmon, some of which are being held by the people. The background shows a lush green forest.



TYPES

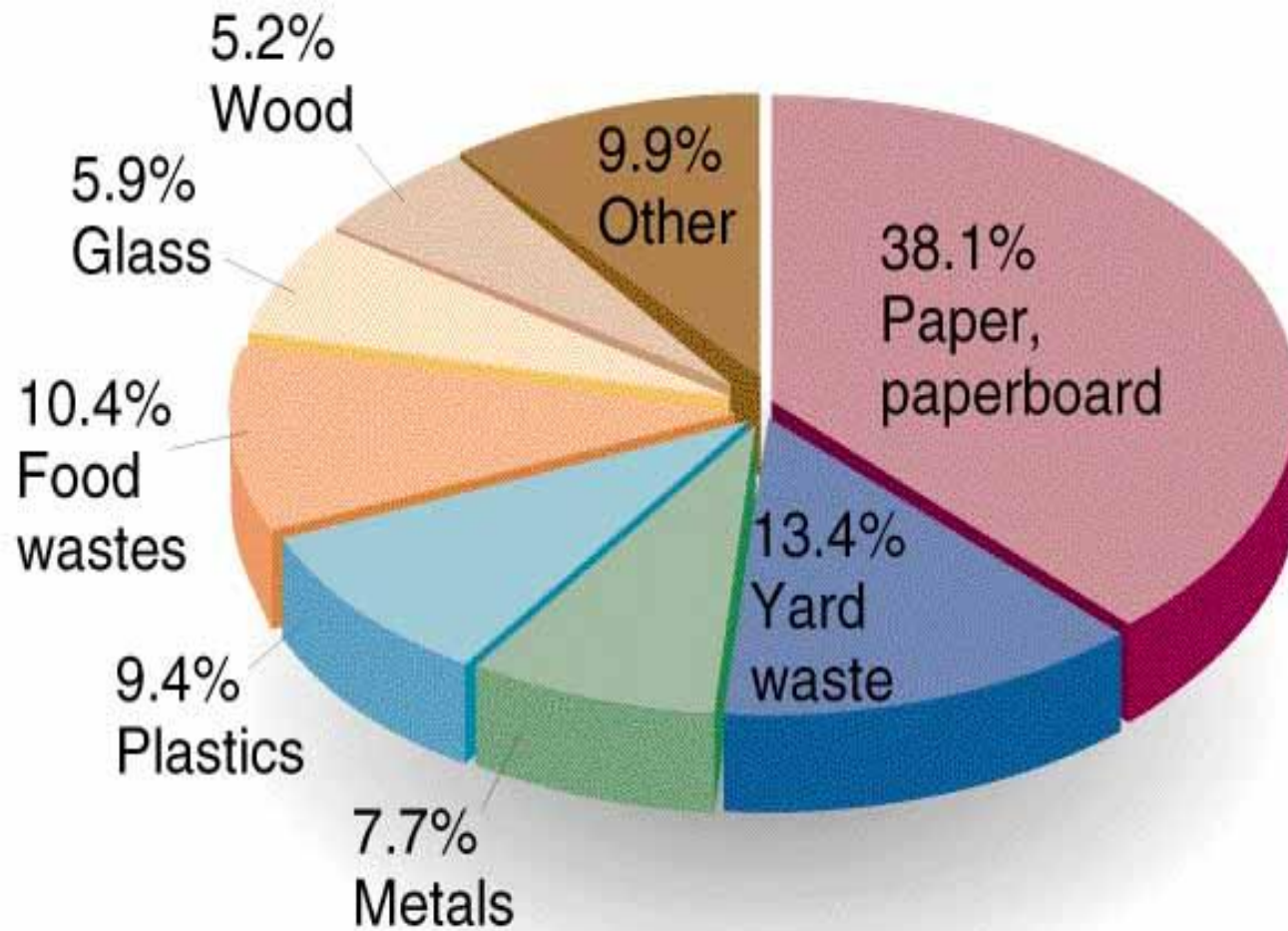
Broadly, there are 3 types of waste, which are as follows

1. Household waste as municipal waste
2. Industrial waste as hazardous waste
3. Biomedical waste or hospital waste as infectious waste

1. MUNICIPAL SOLID WASTE

- Municipal solid waste consist of---
 - household waste
 - construction and demolition debris
 - sanitation residue
 - waste from streets
- With rising urbanization and change in lifestyle and food habits ,the amount of municipal solid waste has been increasing rapidly and its composition changing.

CHARACTERIZATION OF MUNICIPAL SOLID WASTE



HAZARDOUS WASTE

- ◉ Industrial and hospital waste is considered hazardous as it may contain toxic substances
- ◉ Hazardous waste could be highly toxic to humans, animals, and plants. They are
 - corrosive
 - highly inflammable or explosive
- ◉ In the industrial sector, the major generators of hazardous waste are the metal, chemical, paper, pesticide, dye, and rubber goods industries.
- ◉ Direct exposure to chemicals in hazardous waste, such as mercury and cyanide, can be fatal.

HOSPITAL WASTE OR BIOMEDICAL WASTE

- Bio-medical waste means “Any waste which is generated during the diagnosis, treatment or immunization of human beings or animals or in research activities pertaining thereto or in the production or testing of biological”

-Bio-medical waste rules ,1998

- It may include wastes like sharp waste, pathological waste, pharmaceutical waste, genotoxic waste, chemical waste, and radioactive waste etc.

EFFECTS OF SOLID WASTE

A: Health hazard

- If solid waste is not collected and allowed to accumulate, it may create unsanitary conditions.
- This may lead to epidemic outbreaks.
- Many diseases, like cholera, Diarrhea, dysentery, plague, jaundice, or gastrointestinal diseases may spread and cause loss of human lives.

CONTD...

- ◉ In addition, improper handling of solid waste poses a health hazard for workers who come into direct contact with the waste.

B: Environmental impact

- ◉ If solid wastes are not treated properly, decomposition and putrefaction (decay) may occur.
- ◉ The organic solid waste during decomposition may generate obnoxious (intolerable odor)

WASTE MANAGEMENT CONCEPT

- The 3Rs (Reduce, Reuse, Recycle) to be followed for waste management.



REDUCE

- **Disposable goods:** paper plate, paper bowl, Styrofoam cup, plastic spoon, roll of paper towels, paper napkin; Durable goods: ceramic/plastic plate, metal spoon, glass/plastic drinking cup, dish towel, cloth napkin)
- ◎ Recovery of one ton of paper can save 17 trees.

REUSE

- ◉ Instead of buying new containers from the market, use the ones that are in the house.
- ◉ Don't throw away the soft drink can or bottle, cover them with homemade paper or paint on them, and use them as pencil stands or small vases.

RECYCLE

- ◉ Use shopping bags made of cloth or jute which can be used again



What We Cannot RECYCLE



Biscuit and chips packets

Wrappers of bread, noodles and wheat flour (ATTA) bags



Thermocole and aluminium lined dishware

Packaging of diapers and sanitary napkins



Cigarette, pan, 'gutkha' packets, small shampoo sachets

Laminated gift wrapping paper

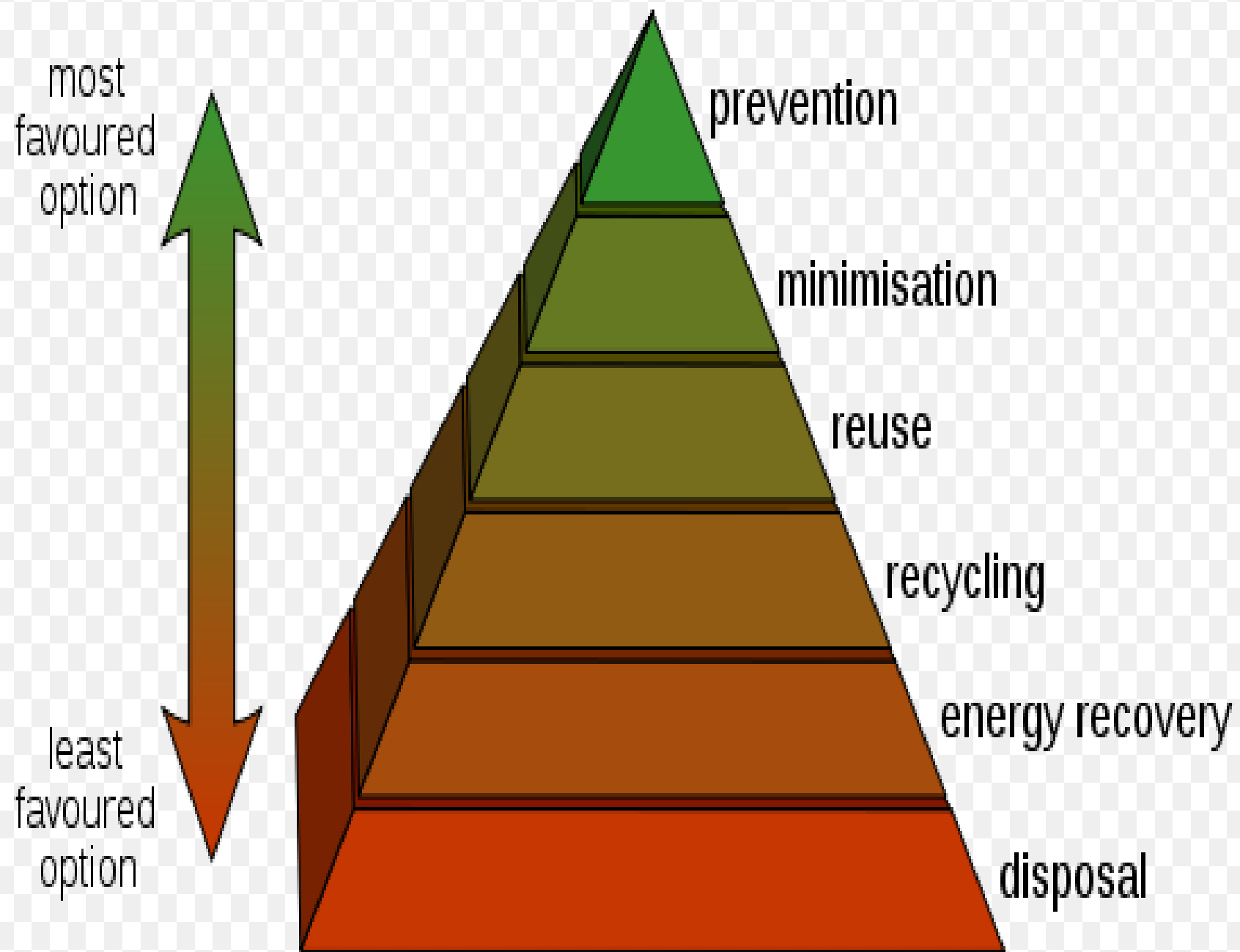


Takeaway Food Parcels

Note: The items listed here have multi layer packaging that makes segregating and recovering plastic or aluminium layers difficult. These items are unlikely to get recycled and end up being a cost to the environment. As far as possible avoid or minimize their use. Look out for or ask for packaging that can be recycled.

WASTE MANAGEMENT HIERARCHY

- ◉ There are a number of concepts about waste management that vary in their usage between countries or regions. Some of the most general, widely used concepts include:
 1. **Waste hierarchy** - The waste hierarchy refers to the "3 Rs" reduce, reuse and recycle, which classify waste management strategies according to their desirability in terms of waste minimization. The waste hierarchy remains the cornerstone of most waste minimization strategies.
 - **The aim of the waste hierarchy is to extract the maximum practical benefits from products and to generate the minimum amount of waste .**



CONTD...

- ◉ **Polluter pays principle** - The Polluter Pays Principle is a principle where the polluting party pays for the impact caused to the environment. With respect to waste management, this generally refers to the requirement for a waste generator to pay for appropriate disposal of the unrecoverable material.

SOLID WASTE MANAGEMENT

Waste management is the

- ◉ storage
- ◉ collection
- ◉ transport and handling
- ◉ recycling
- ◉ disposal and monitoring of waste materials.

STORAGE

◉ Storage:

- Galvanized steel dust bin
- Paper sack
- Public bins



COLLECTION

◉ Collection

- House-to-house collection
- Collection from the public bins



WASTE HANDLING AND TRANSPORT

- ◉ Waste handling and separation involve activities associated with waste management until the waste is placed in storage containers for collection. Handling also encompasses the movement of loaded containers to the point of collection.
- ◉ waste is transferred from a smaller collection vehicle to larger transport equipment



RECYCLING

- Recycling refers to the collection and Refuse of waste materials such as empty beverage containers.
- The materials from which the items are made can be processed into new products.
- Recycling materials may be collected separately from general waste using dedicated bins.



METHOD OF DISPOSAL

1. Dumping
2. Controlled Tipping or Sanitary Landfill
3. Incineration
4. Composting
5. Manure pits
6. Burial

OBJECTIVES

- ⦿ Public hygiene and health.
- ⦿ Reuse, recovery and recycle
- ⦿ Energy generation
- ⦿ Sustainable development
- ⦿ Aesthetics

1. DUMPING

- ◉ Low-lying areas.
- ◉ Mainly for dry refuse
- ◉ method and reclaimed land given for cultivation.
- ◉ Unsanitary method
 - Exposed to flies and rodents
 - Nuisance
 - Dispersed by wind
 - pollution of surface water



2. CONTROLLED TIPPING/ SANITARY LANDFILL

- ⦿ Satisfactory method
 - Material placed in a trench
 - compacted with earth at the end of the working day.
- ⦿ Modified sanitary land fill-where compaction and covering are accomplished once or twice a week.

◉ 3 Methods

1. Trench method
2. Ramp method
3. Area method

Refuse is compacted on its exposed surface with excavated earth (30 cm).

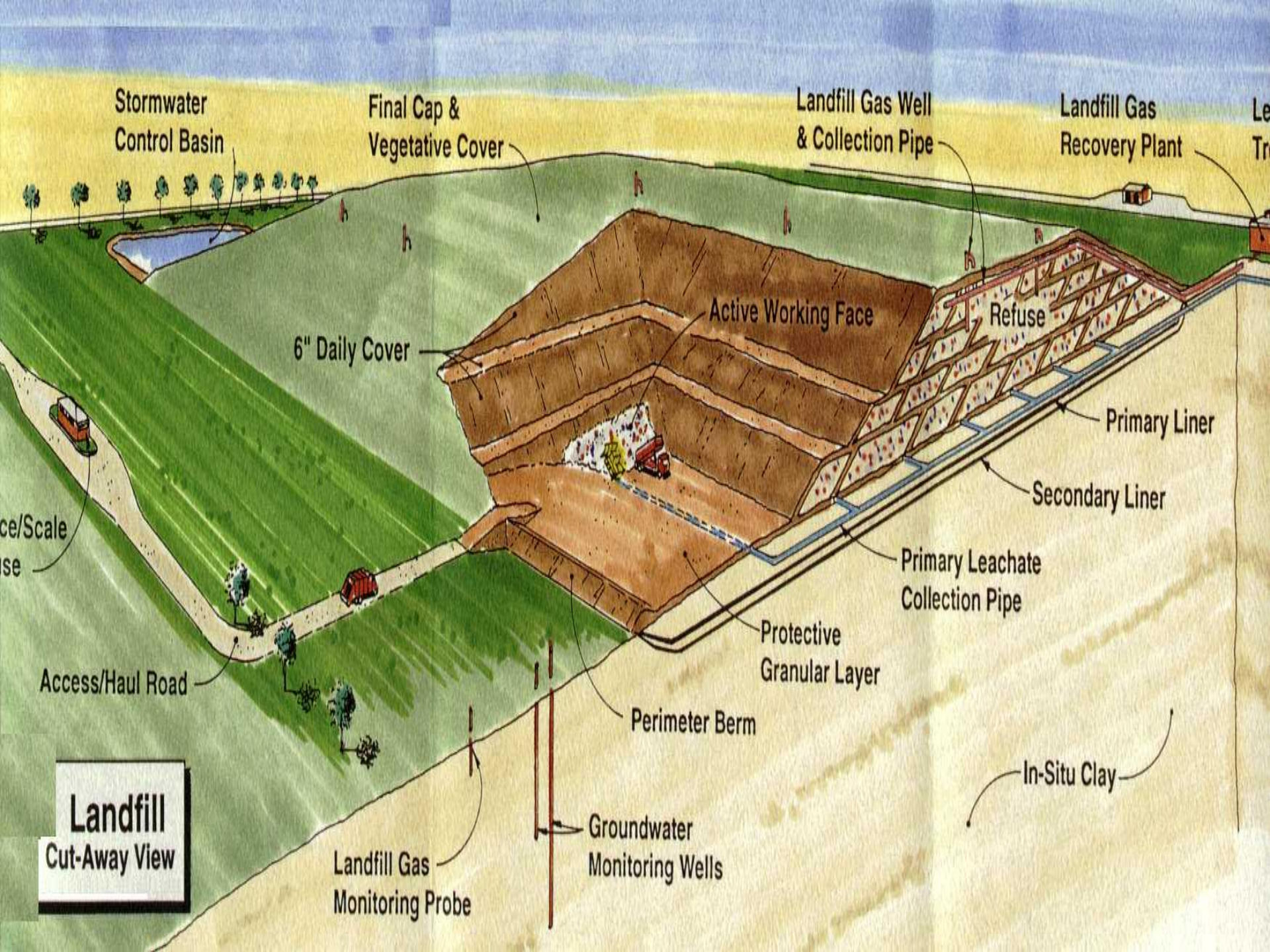
TRENCH METHOD

- ◉ Long trench of 6-10 feet deep and 12-36 feet wide.
- ◉ Refuse is compacted and covered with excavated earth.
- ◉ Refuse is filled up to 6 feet.
- ◉ It is estimated one acre of land per year for 10,000 population.
- ◉ **RAMP METHOD:** suited where the terrain is moderately slopping.

AREA METHOD

- ◉ Used for filling land depressions, disused quarries and clay pits.
- ◉ Refuse is deposited, packed and consolidated in uniform layers for 6-8 feet.
- ◉ Each layer is sealed with a mud cover at least 12 inches.
- ◉ Sealing prevents infestation by flies and rodents.
- ◉ Prevents nuisance of smell and dust.

- ◉ Changes
 - Chemical
 - Bacteriological
 - Physical
- ◉ The temperature rises to over 60 deg. C within 7 days and kills all pathogens and hastens the decomposition process.
- ◉ It takes 4 to 6 months for complete decomposition.



Stormwater
Control Basin

Final Cap &
Vegetative Cover

Landfill Gas Well
& Collection Pipe

Landfill Gas
Recovery Plant

Le
Tr

6" Daily Cover

Active Working Face

Refuse

Primary Liner

Secondary Liner

Primary Leachate
Collection Pipe

Protective
Granular Layer

Perimeter Berm

In-Situ Clay

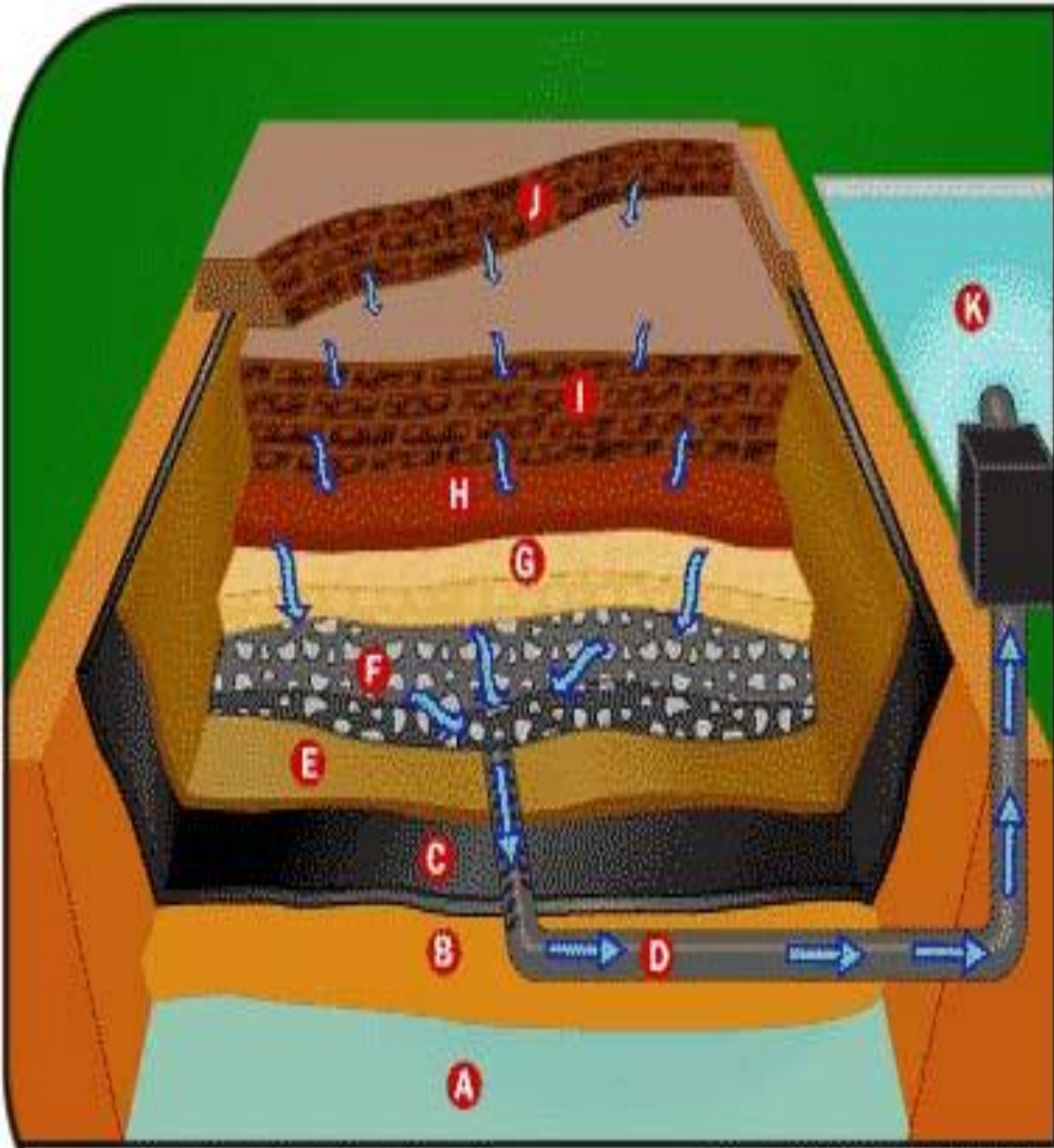
Groundwater
Monitoring Wells

Landfill Gas
Monitoring Probe

Access/Haul Road

ce/Scale
se

Landfill
Cut-Away View



- A** Ground Water
- B** Compacted Clay
- C** Plastic Liner
- D** Leachate Collection Pipe
- E** Geotextile Mat
- F** Gravel
- G** Drainage Layer
- H** Soil Layer
- I** Old Cells
- J** New Cells
- K** Leachate Pond



SINKHOLE



3. INCINERATION

- ◉ It is a disposal method in which solid organic wastes are subjected to combustion to convert them into residue and gaseous products.
- ◉ This process reduces the volumes of solid waste to 20-30% of the original volume.
- ◉ Also described as thermal treatment
- ◉ Where land is not available
- ◉ Hospital waste



4.COMPOSTING



- ◉ Method of combined disposal of refuse and night soil/ sludge
- ◉ Principal byproducts are CO_2 , Water and heat
- ◉ End product- compost
- ◉ Methods
 - a. Bangalore method
 - b. Mechanical composting
 - c. Vermicomposting

Raw materials

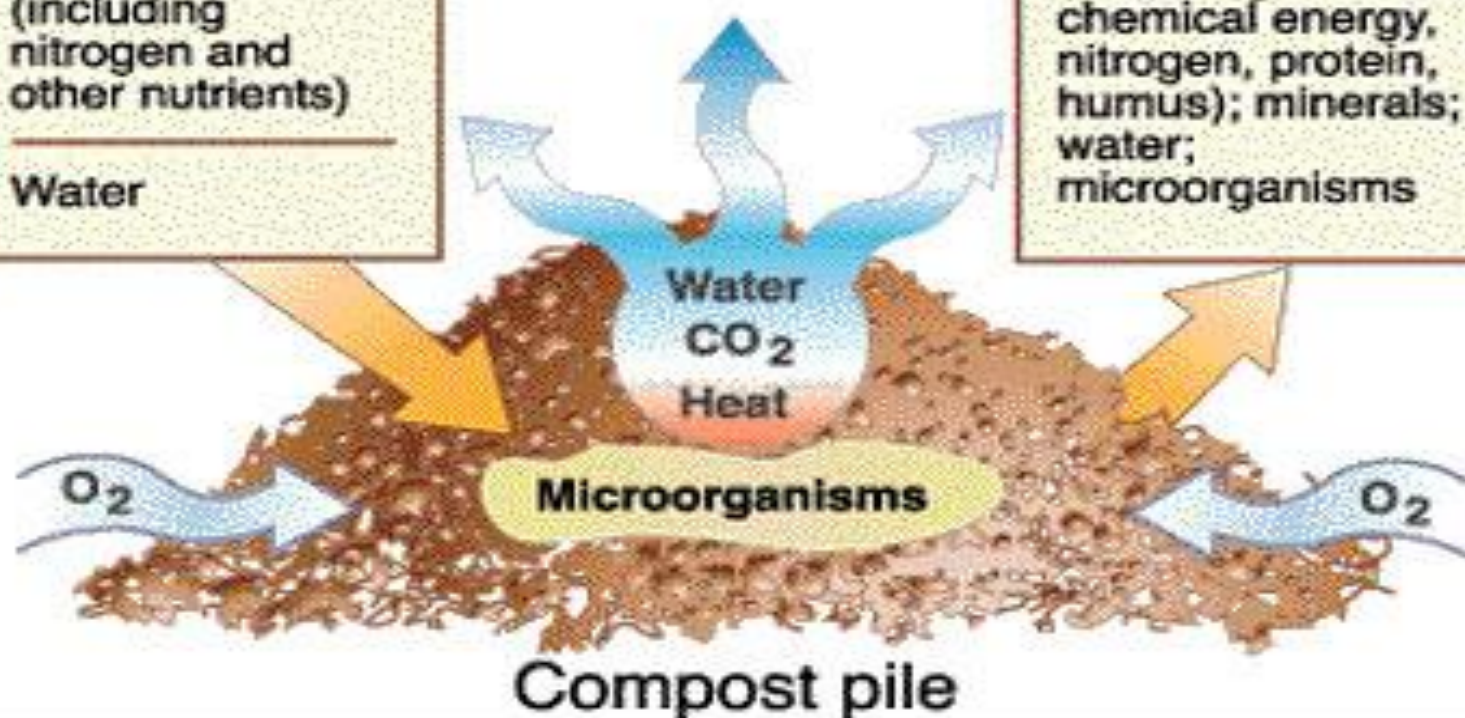
Organic matter
(including carbon,
chemical energy,
protein, nitrogen)

Minerals
(including
nitrogen and
other nutrients)

Water

Finished compost

Organic matter
(including carbon,
chemical energy,
nitrogen, protein,
humus); minerals;
water;
microorganisms



A. HOT FERMENTATION PROCESS/ ANAEROBIC METHOD

- ◉ IISc- Indian Council of Agricultural Research.
- ◉ Trenches are dug 3ft deep, 5-8ft broad, 15-30ft long.
- ◉ Composting procedure
 - 1. Layer of Refuse- 15 cm
 - 2. Layer of Night soil - 5 cm

Physical, chemical and biological changes takes place

B. MECHANICAL COMPOSTING (AEROBIC METHOD)

- ◉ Compost manufactured by processing raw materials.
- ◉ 1. Screening
- ◉ 2. Pulverization (<2inches)
- ◉ 3. Mixing
- ◉ 4. Incubation

Process completed in 4-6 week

C: vermicomposting :It is a method of disposal of kitchen *and* plate wastes, which serves the dual purpose of disposing off the garbage as well as proving eco-friendly.



5. MANURE PITS



- ◉ Mostly used in rural areas
- ◉ Digging “manure pits” is to prevent the refuse from being thrown around the houses.
- ◉ The garbage, cattle dung, straw, and leaves should be dumped into the manure pits and covered with earth.
- ◉ Two pits will be needed
- ◉ In 5-6 months, the refuse is converted into manure, which can be returned to the field.

6. BURIAL



- ◉ Suitable for small camp
- ◉ A trench 1.5m wide & 2 m deep is excavated
- ◉ The refuse is covered with 20 -30cm of earth
- ◉ When the level in the trench is 40cm from ground level, the trench is filled with earth & compacted
- ◉ 4-6 months

TECHNOLOGY

- Technologies like

RFID(Radio frequency identification) tags are now being used to collect data on presentation rates for curbside pick-ups.

GPS tracking is particularly evident when considering the efficiency of ad hoc pick-ups (like skip bins or dumpsters), where the collection is done on a consumer request basis.

Rear vision cameras are commonly used for OH&S (Occupational Health & Safety) reasons, and video recording devices are becoming more widely used, particularly concerning residential services.

ZERO WASTE SYSTEM

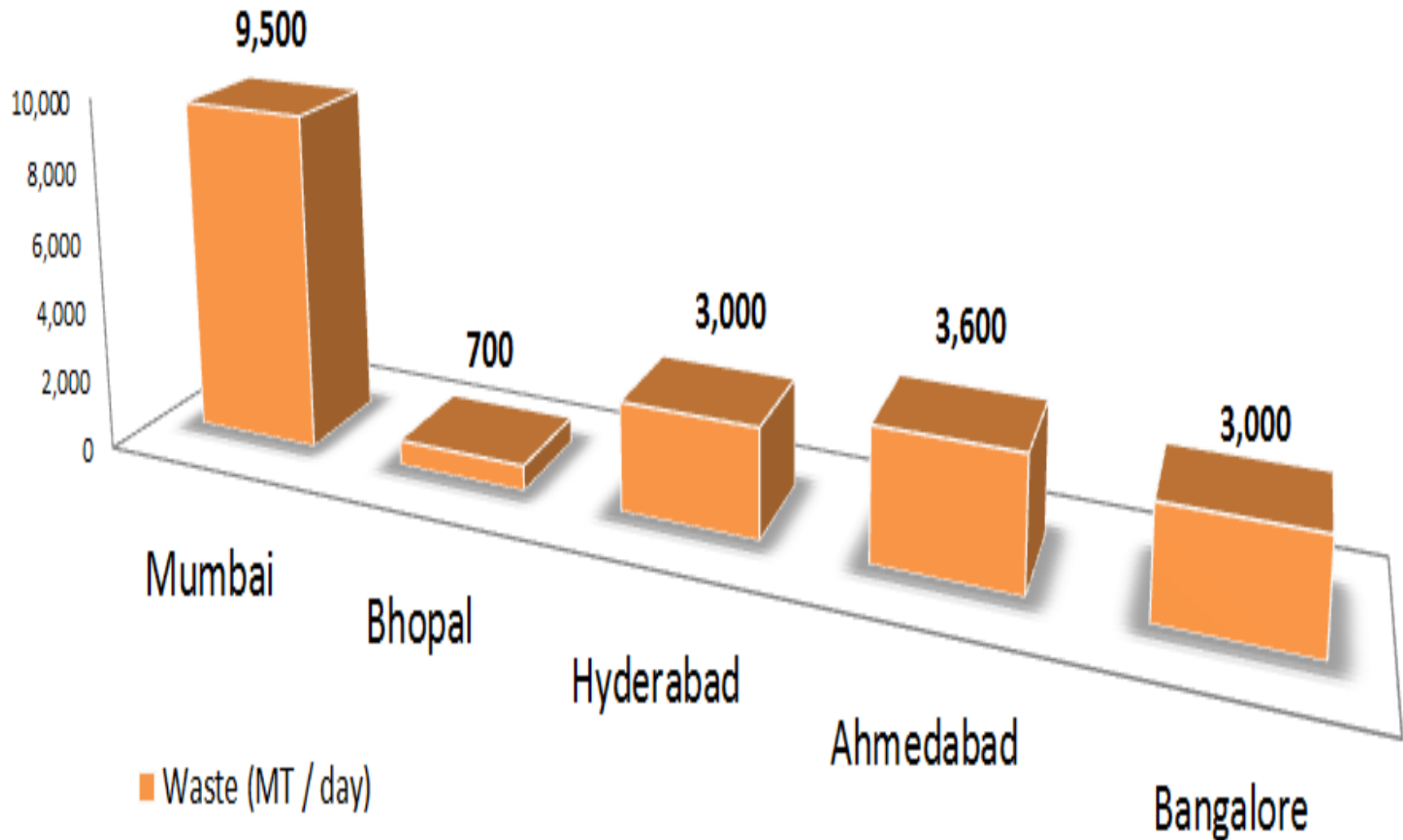
- ◉ A zero-waste system, which was founded by a PhD chemist. Paul Parmer in Oakland.
- ◉ It is a philosophy that encourages the redesign of the resource life cycle so that all products are reused.



NOW WE CAN ACHIEVE
ZERO WASTE

RECYCLING ALL OUR HOUSEHOLD WASTE
IS NOW POSSIBLE

WASTES PER INDIAN CITIES



WASTE MANAGEMENT PLANT IN MYSORE

- There are two solid waste management plant in Mysore.

J.P Nagar
vidyaranyapuram



Mysore gets two more waste treatment plants

H M Aravind | TNN

Mysore: The city will get two more solid waste management plants for better disposal of the garbage generated daily at the tourist hub. Plans to shift the existing plant to the outskirts have been dropped.

As of now, of the 402 tonnes of garbage generated daily, 200 tonnes is processed every day and the rest is dumped at sewage-treatment plant in Vidyaranyapuram, leading to pollution and complaints from the residents in the area.

The two additional plants will be built at Kesare and Rayanakere towards the western and eastern sides of the city. Each of these plants will have a capacity to process 200 tonnes per day. The Mysore City Corporation budgeted Rs 5 crore for the project.

Though the local body had announced to shift the centralized solid waste management plant at Vidyaranyapuram to Rayanakere, it has been shelved. The city's oldest plant will continue to oper-

ate but the dependence on it will be reduced gradually.

Announcing the plan, taxation and finance standing committee chairman B L Bhyrappa on Tuesday said: "There is a need to build additional plants to cut down pressure on the existing plant and handle additional generation of garbage in future.

Mysore scored over Indian cities bagging the second cleanest city award after Chandigarh by the Union urban development ministry recently. Bhyrappa told TOI the plan is to divide the city's 65 wards into three parts and process the garbage at the plant near them.

The MCC, which is setting up four additional zero solid waste management plants, set aside Rs 50 lakh and Rs 1 crore to upgrade the five plants. With the MCC claiming to have achieved 60% segregation of waste at the source, it is focusing on achieving 100% segregation to better manage the garbage. It will take up education programmes with a budgetary allocation of Rs 60 lakh



FULL COVERAGE: P 2

BBMP(BRUHAT BANGALORE MAHANAGAR PALIKE)

- ◉ BBMP represents the third level of government & forth largest municipal corporation in India .
- ◉ It is responsible for the development of the city , health hygiene, licensing trade & education.



ಬೃಹತ್ ಬೆಂಗಳೂರು ಮಹಾನಗರ ಪಾಲಿಕೆ

Bruhat Bengaluru Mahanagara Palike

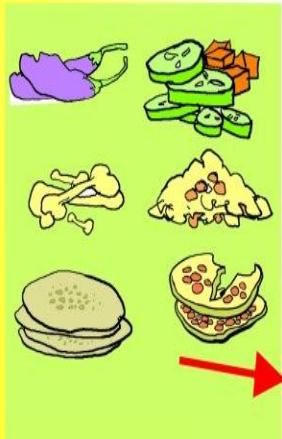
From October 1st onwards, BBMP will collect
only **WET** and **SANITARY** waste **DAILY**

DRY waste will be collected **WEEKLY**



Wet Waste

- ❖ cooked food (veg / non-veg)
- ❖ uncooked food (veg / non-veg)
- ❖ fruit and flower waste
- ❖ fallen leaves and similar compostable materials



Dry Waste

- ❖ paper / plastic / wood
- ❖ metals / fabrics / cloth rags
- ❖ glass / wire / leather
- ❖ rexin / rubber
- ❖ thermocol etc...



wet waste
in polythene
not accepted



Sanitary Waste
wrap in newspaper
marked with a red cross



Used Sanitary Napkins Used Child /Adult Diapers Used Bandage/Band Aids Used Ear Buds



Contact BBMP control room no : 080-2266 0000
e-mail : swm.bbmp@gmail.com website : www.bbmp.gov.in
f : [facebook/bbmp-speak-on-solid-waste-management](https://www.facebook.com/bbmp-speak-on-solid-waste-management)

Bruhat Bangalore Mahanagara Palike

Join hands to create a garbage-free Bangalore

WASTE MANAGEMENT POLICIES

- ◉ Environment protection act, 1986
- ◉ Hazardous waste rule 1989
- ◉ Bio-medical waste rule 1998
- ◉ Municipal solid waste rule 2000
- ◉ Waste management act 1996
- ◉ Solid waste policy in India 2006

RECOMMENDATION

- ◉ The improvement of people and private sector through NGOs could improve the efficiency of solid waste management.
- ◉ Public awareness should be created especially at primary level.
- ◉ Littering of solid waste should prohibited in cities towns and urban areas.
- ◉ Moreover, house to house collecting solid waste should be .

CONTD...

- ◉ The collection bins must be have a large enough capacity to accommodate 20% more than the expected waste generation in the area.
- ◉ Municipal authorities should maintain the storage facilities to avoid unhygienic & unsanitary condition.
- ◉ It is advisable to move from open dumping to sanitary land filling in a phased manner.

REFERENCE

- ◉ Park K. Park's Textbook of Preventive and Social Medicine. 22nd ed. Jabalpur: Banarsidas Bhanot Publishers; 2013. Chapter 20, Communication for Health Education; p699-701
- ◉ Bhalwar, Rajvir et al, 2009, *Textbook of Public Health and Community Medicine*, Prune, WHO/AFMC;p715-717
- ◉ BK Mahajan. (2011) Textbook of preventive and social medicine. 7th Edition. Jaypee Brothers Medical Publishers (P) Ltd., Daryaganj, New Delhi.;p71-73
- ◉ AH Suryakantha. Community Medicine With Recent Advances. 3rd Edition. Jaypee Brothers Medical Publishers (P) Ltd., Daryaganj, New Delhi; 2013;p233-237



T
H
A
N
K

YOU